

SPID: a spectral partial information decomposition framework for quantifying information about cognitive variables in oscillatory brain networks

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Abstract

Neural oscillations are thought to play a central role in encoding and transmitting cognitive information across large-scale brain networks, yet the relative contributions of phase synchrony and amplitude co-modulations to distributed coding remain unclear. A key obstacle is the absence of tools that can simultaneously quantify task-relevant information in the frequency domain and disentangle its phase and amplitude components across pairwise and higher-order interactions. Here, we introduce Spectral Partial Information Decomposition (SPID), an approach that combines frequency-domain mutual information with Partial Information Decomposition to disentangle oscillatory interactions into unique, redundant, and synergistic information about cognitive variables. We validated SPID on Kuramoto and Stuart–Landau oscillator networks, including a whole-brain model constrained by macaque anatomical connectivity; SPID accurately recovers ground-truth encoding schemes and reveals that phase relations and amplitude co-modulations act as complementary coding channels with both redundant and synergistic components. SPID further extends this decomposition to higher-order functional interactions, beyond pairwise relations, enabling the characterization of how cognitive information is collectively distributed across multiple oscillatory edges via redundant and synergistic encoding. To illustrate biological applicability, we applied SPID to local field potentials (LFPs) recorded from the macaque fronto-parietal network during a working memory task. In this dataset, SPID identified beta-band amplitude co-modulations as the primary carrier of stimulus information, and revealed that higher-order phase interactions exhibit both redundant and synergistic structure during the memory delay. These results establish SPID as a principled tool for dissecting the informational architecture of oscillatory brain networks.

INTRODUCTION

A central hypothesis in neuroscience posits that cognitive functions emerge from the coordinated activity and interactions of widespread neural populations (Varela et al., 2001b; Bressler and Menon, 2010) resulting in self-organized networks encoding and routing information relevant for cognitive functions (Battaglia and Brovelli, 2020; Deco et al., 2015). An increasing number of studies indicate that neural functional interactions, defined as statistical relationships between the activity of different brain regions (Panzeri et al., 2022), reflect task-relevant information (Panzeri et al., 2022; Combrisson, 2022), with evidence that this information is structured into redundant and synergistic components (Varley et al., 2023a; Luppi et al., 2024; Combrisson et al., 2024, 2025). Redundancy captures the shared information about cognitive variables that is contained in both brain regions, or neuronal populations individually. In contrast, synergy reflects the additional information present only in the joint observation of both regions, which cannot be obtained from either one individually. Current hypotheses suggest that redundant information may enhance robustness by ensuring that critical information is consistently duplicated and distributed to downstream readout (Valente, 2021; Combrisson et al., 2024). Synergistic information may be associated with flexibility and provide functional advantage, because it would enable the full exploitation of possible combinations of neurons or brain areas, making their informational capacity to scale exponentially with the system size (Rosas et al., 2020). Indeed, synergistic interactions have been reported between distant associative regions during high-level cognition (Luppi et al., 2022; Varley et al., 2023a; Combrisson et al., 2024, 2025) and, at the microscale, in populations of neurons within a cortical column of the visual cortex and across areas of the visuomotor network (Nigam et al., 2019).

Additionally, several studies suggest that task-relevant information is contained in collective and higher-order statistical dependencies across multiple neurons or brain regions, beyond pairwise relations (Martignon, 2000; Yu, 2011; Shahidi et al., 2019; Chelaru, 2021; Panzeri et al., 2022; Varley et al., 2023b; Santoro et al., 2023). Generally, structural higher-order interactions are at the core of complex systems (Crutchfield, 1994; Battiston et al., 2020; Battiston, 2021) and they are thought to support collective behavior in social systems (Benson et al., 2016), ecology (Grilli et al., 2017; Levine et al., 2017) and biology (Sanchez-Gorostiaga et al., 2019). In brain dynamics, higher-order statistical dependencies among multiple neurons or brain regions may reflect complex coordinated activity associated with emergent functional properties, such as information integration and cognitive flexibility (Schneidman et al., 2006; Köster et al., 2014; Chelaru, 2021; Santoro et al., 2024). Indeed, a recent study has shown that goal-directed learning variables, such as information gain, are reflected in synergistic and higher-order functional interactions between distant brain areas, and that this information is directionally structured toward key brain structures such as the prefrontal reward circuits (Combrisson et al., 2025). Taken together, the tradeoff between synergistic and redundant information within high-order functional interactions may support an efficient and distributed

representation of information relevant to cognitive functions.

A large body of evidence also suggest that neural oscillations play a crucial role in supporting brain-wide interactions and cognitive processes, such as selective attention (Womelsdorf and Fries, 2007; Womelsdorf et al., 2007; Bosman et al., 2012; Bonnefond et al., 2017; van Kerkoerle et al., 2014; Fries et al., 2001) and working memory (Salazar et al., 2012; Miller et al., 2018; Lundqvist et al., 2016; Dotson et al., 2018). Several accounts have been proposed to explain the mechanisms supporting neural interactions, including phase synchronization between population-level activities (Singer and Gray, 1995; Engel et al., 2001; Varela et al., 2001b; Fries, 2015, 2009; Buzsáki and Draguhn, 2004; Siegel et al., 2012), frequency-specific feedforward and feedback influences (Vezoli et al., 2021; Bastos et al., 2015; Xiong et al., 2024), and ~~from~~ entrainment ~~or~~ [via the](#) resonance between the sender and receiver ~~integrating~~ oscillatory ~~and~~ [or](#) transient dynamics (Vinck et al., 2023, 2025). Although these alternative accounts differ in their specific mechanisms, they all highlight the importance of phase relations between neural oscillations and co-modulations [of](#) transient dynamics. However, progress toward a unified interpretation is limited by the lack of tools to quantify information about cognitive variables in oscillatory neural networks, and to disentangle the role of phase and amplitude relations.

The aim of the [present](#) study [is](#) to provide measures quantifying redundant and synergistic information about cognitive variables contained in pairwise and higher-order interactions, and [to](#) disentangle information reflected in transient co-modulations of oscillatory power and phase relations across frequency bands. To do so, we combined information theoretical metrics based on frequency-domain mutual information (Brillinger and Guha, 2007; Malladi et al., 2018) and partial information decomposition (Williams and Beer, 2010; Wibral et al., 2017; Lizier et al., 2007). We first validated our approach through simulations based on the Kuramoto and Stuart–Landau models of coupled oscillators, both pairwise and higher-order interactions. The former is a classical model for studying oscillatory networks, whereas the latter extends it by incorporating both phase and amplitude, providing a minimal system capable of exhibiting subcritical Hopf bifurcations as well as transient and steady-state oscillations (Ponce-Alvarez and Deco, 2024). We then tested the methods on simulations of multiple oscillators coupled according to macaque monkey structural connectivity (Markov et al., 2014). Such model captures inter-areal interactions through synchronization and modulation of oscillatory power. We finally provided proof-of-concept on local field potentials (LFPs) recorded from macaque cortical regions during a working memory task (Salazar et al., 2012), demonstrating that our approach is able to provide key information about the cognitive variables through phase- and amplitude-based interactions.

RESULTS

We present a novel information decomposition approach for oscillatory networks. The methodology quantifies information about cognitive variables contained in oscillatory interactions by combining frequency-domain mutual information (Malladi et al., 2018; Brillinger and Guha, 2007) and the partial information decomposition framework (Williams and Beer, 2010; Wibral et al., 2017). The methodology quantifies the information contained in oscillatory interactions about task-related factors, such as sensory, decision or cognitive factors. It separates information contained in power modulations or phase differences between neural signals in the frequency domain, and it disentangles redundant and synergistic information contained in pairwise up to higher-order interactions. We tested the methodology on ground-truth neural interactions generated using oscillatory network models encoding task variables via pairwise and higher-order interactions. We then tested the methods on oscillatory activity generated using dynamical models based on realistic large-scale anatomical connectivity from non-human primates (Markov et al., 2014). Finally, we provide a proof-of-concept on simultaneously-recorded local field potentials (LFPs) recorded from the fronto-parietal network of macaque monkeys performing a working memory task (Salazar et al., 2012).

Information about cognitive variables in pairwise oscillatory interactions

We simulated the modulation of neuronal interactions through changes in neuronal synchronization and coupling strength in a two-node network (pairwise interaction). This may correspond, for example, to a top-down attentional effect modulating the functional connectivity between two brain regions (e.g., V1 and V4) by transiently improving phase synchrony through task-specific increases in coupling strength (Fries, 2005; Buschman and Miller, 2007). The modulation in phase synchrony would parallel the strength of attentional modulation, enabling phase-based information about cognitive variables (Womelsdorf et al., 2007).

We simulated a two-node Kuramoto model with a central oscillation frequency of 40 Hz and unidirectional coupling from X_i to X_j . As a reminder, the Kuramoto model describes the phase dynamics of coupled oscillators assuming fixed amplitudes, it therefore allows only phase encoding. To simulate a task-related modulatory effect of the phase relationship between the two nodes both in time and trials, we varied the coupling strength $g(t)$ (see Methods). Within trial, the coupling strength followed a Gaussian profile (Fig. 1B), thus producing transient phase synchronization in the time interval from 0.1 to 0.3 s. Across trials, the amplitude of this Gaussian profile was varied from 1 to 100 (over $n = 1000$ trials), thereby producing an encoding of the task-related variable Y into the phase relationship between X_i and X_j . Panel 1C shows exemplar time series for the two nodes, illustrating the clear synchronization during the stimulus interval. Consistently, the Kuramoto order parameter (KOP) exhibited a transient increase during the input period (Fig. 1D).

Time-frequency representation of spectral coherence between nodes was computed using

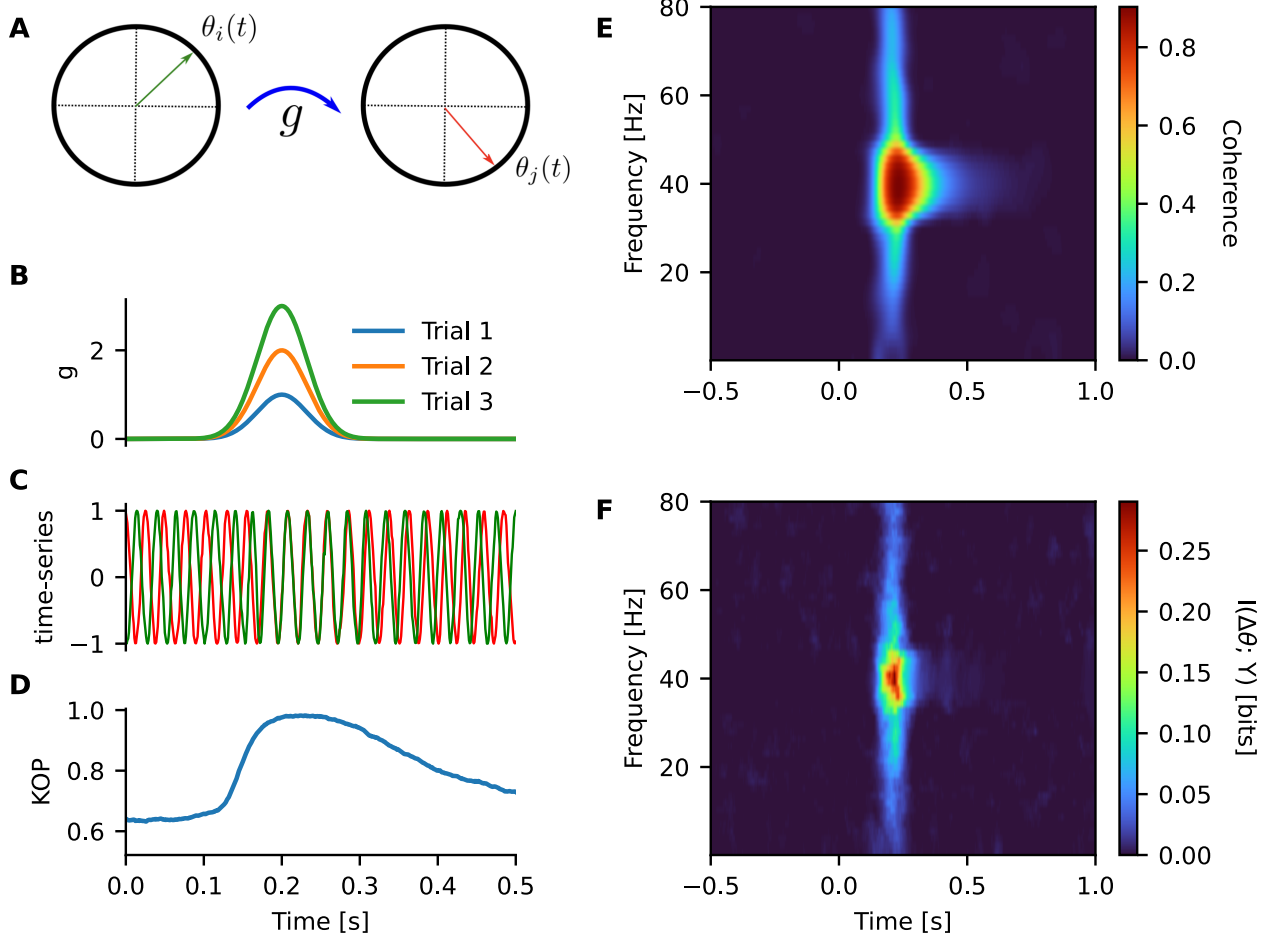


Figure 1: **Phase-based encoding in pairwise neural interactions.** (A) Simulation of two Kuramoto oscillators (see Eq. 13; using identical parameters for both nodes: $\omega = 2\pi 40$ Hz, $\eta = 2.5$, and coupling strength $C_{12} = 10$) coupled via an undirected edge. The time series of the simulated dynamical system are represented by their instantaneous phases $\theta(t)$ (depicted by green and red arrows in the unit circle). The coupling strength $g(t)$ is unidirectional from X_i to X_j . (B) The coupling strength $g(t)$ between the two nodes was time-modulated such that it was transiently non-zero from 0.1 to 0.3 s, following $g(t) = 8Y/(2\pi f)$, where Y was the maximum coupling amplitude, increased from 1 to 100 across trials ($n = 1000$). (C) Time series of the simulated Kuramoto nodes for the trial with the strongest coupling. (D) Kuramoto order parameter (KOP) time series averaged across trials. (E) Time-frequency coherence spectrogram between the two nodes computed using wavelet decomposition ($n_{\text{cycles}} = f_c/4$; for $f_c \in [0.1, 80]$ Hz). (F) Mutual information between the phase difference of the two nodes (computed from the wavelet coefficients according to Eq. 10) and the coupling amplitudes Y .

128 wavelet transform methods, revealing a pronounced transient synchronization at 40 Hz correspond-
 129 ing to the input (Fig. 1E). We then computed the mutual information in the time and frequency
 130 domains between the phase difference of X_1 and X_2 ($\Delta\theta$) and the task variable Y , which reflected
 131 the across-trial modulation of coupling amplitude. The resulting spectrum showed a clear peak
 132 at 40 Hz, demonstrating that the modulation of coupling strength was contained in the phase
 133 relationship between the two nodes (Fig. 1F). Overall, the results provide evidence that the pro-

posed methodology successfully captures the amount of information contained in phase relations between nodes about a task-related variable modulating coupling strength.

Information about cognitive variables in higher-order phase relations

We then evaluated the methodology's ability to quantify the information contained in higher-order oscillatory interactions. The rationale was to generalize the previous analysis to account for mutual information between two or more phase differences and the task variable Y . To do so, we employed the Partial Information Decomposition (PID) framework, which provides a mathematical formalism for the decomposition of multivariate mutual information between a set of predictors and a target variable into unique, redundant and synergistic information terms (Williams and Beer, 2010; Wibral et al., 2017). In the current study, the PID formalism was used to decompose the total interdependence between two phase differences between nodes $\Delta\theta_1$ and $\Delta\theta_2$, respectively, and the task variable modulating coupling strength Y , into redundancy and synergy terms. The redundant information captures the portion of task-related information that is shared across all phase-difference pairs, that is, information redundantly represented across edges. In contrast, the synergistic information reflects information that emerges only when considering the joint activity of two edges, beyond what any single edge contributes individually. Together, these measures indicate whether task-related information in a network of interacting edges is organized redundantly (shared) or synergistically (complementary) across phase relations.

We simulated modulation of neuronal interactions through changes in neuronal synchronization and coupling strength in a network with three nodes using a network of Kuramoto oscillators. Generally, a chain structure promotes redundant encoding, because information is propagated sequentially, causing downstream nodes to inherit overlapping representations from upstream sources, whereas synergistic encoding requires convergent integration (i.e., colliders) of independent inputs. To simulate a higher-order oscillatory interaction in which phase relations contain task-related information with a greater redundant (shared) than synergistic component, we therefore simulated a feedforward Kuramoto network model organized in a chain structure (Fig. 2A). The coupling strength varied across trials, and in a trial-based and time-varying manner across all edges. As expected, the time-frequency representation for the chain of oscillators showed stronger redundant than synergistic information (Fig. 2B). We then simulated a higher-order oscillatory interaction dominated by synergistic encoding using a collider structure (Fig. 2C). This structure produced a relatively stronger synergistic information as evidenced by the peak value in the time-frequency representation (Fig. 2d). Finally, we simulated the oscillatory activity of a network of three visual cortical areas (V1, V2 and V4 circuit) connected anatomically accordingly to retrograde tracing data, the so-called fraction of labeled neurons (FLN) (Markov et al., 2013). Within such recurrent structure, task information produced both redundant and synergistic information contained in higher-order phase relations (Fig. 2F). To conclude, our simulations demonstrate that redundant and synergistic information about cognitive variables can be disentangle for higher-

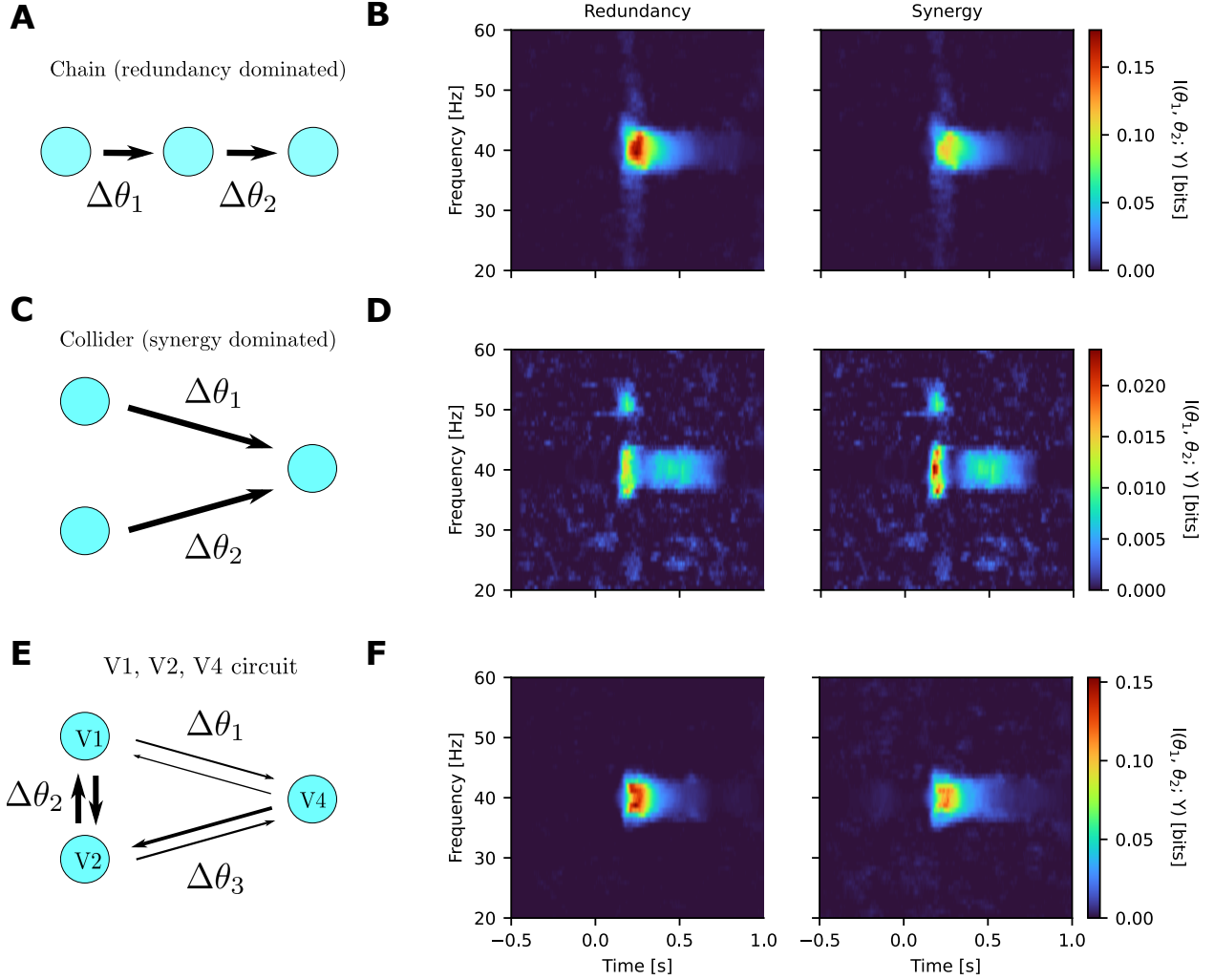


Figure 2: Phase-based information in higher-order neural interactions. Simulation of three Kuramoto oscillators (Eq. 13) with the same parameters for all nodes ($\omega = 2\pi 40$ Hz; $\eta = 1.5$), coupled via unidirectional edges. The coupling strength among nodes was transiently modulated from 0.1 to 0.3s using a Gaussian profile with coupling amplitude increasing from 1 to 100 across trials ($n = 1000$). **(A)** Redundant encoding was simulated using a chain structure. Phase differences between nodes $\Delta\theta_1$ and $\Delta\theta_2$ were derived from the Wavelet coefficients of the nodes' time-series (Eq. 10; $n_{\text{cycles}} = f_c/2$; for $f_c \in [20, 60]$ Hz). **(C)** Synergistic encoding was simulated using a collider structure. **(E)** Simulation of a three-nodes Kuramoto network structured in a recurrent connectivity. Structural connectivity strengths are proportional to their anatomical connectivity measured via retrograde tracing (FLN; Markov et al. (2013)) between three hypothetical visual cortical areas V1, V2 and V4 circuit. The effective connectivity between nodes i and j is given by $C_{ij} \cdot \text{FLN}_{ij}$, and the arrows representing edges are scaled accordingly. **(B, D and F)** Time–frequency representation of redundant (left panel) and synergistic (right panel) information contained in phase differences $\Delta\theta_1$ and $\Delta\theta_2$ about the task-related variable Y , computed using the PID approach (Eqs. 6 and 7).

171 order interactions, therefore opening the possibility to study the distributed encoding of cognitive
 172 variables in oscillatory networks via phase relations.

Dissociating information about cognitive variables in phase relations and amplitude co-modulations in oscillatory networks

Recent hypotheses suggest that aperiodic transients and oscillatory dynamics may differentially contribute to neural interactions along feedforward and feedback pathways for sensory inference and predictive processing (Vinck et al., 2023, 2025). Within this framework, neural oscillations are proposed to contain and stabilize neural representations over time and facilitate plasticity processes during the late, feedback-dominated phases of sensory processing. Transients and non-oscillatory dynamics are thought to constitute the primary mechanism for inter-areal communication during sensory inference. Dissociating the contribution of phase relations of oscillatory dynamics and power modulations of transients is therefore important for testing such hypotheses. To address this issue, we simulated a two-node network based on Stuart-Landau oscillators, instead of Kuramoto model. Indeed, whereas the Kuramoto model describes the phase dynamics of coupled oscillators assuming fixed amplitudes, the Stuart-Landau (SL) model describes both amplitude and phase dynamics, allowing oscillations to emerge, grow, or decay. We simulated task-relevant information by modulating the coupling from X_i to X_j as a Gaussian input of varying amplitude across trials, similar to previous simulations based on Kuramoto oscillators. However, contrary to the Kuramoto model, the SL model does not produce single-frequency **exclusively phase-synchronization encoding**, because SL oscillators have both amplitude and phase degrees of freedom. Modulation of $g(t)$ drives both amplitude and phase modulation of the receiving oscillator, therefore producing sidebands in the frequency spectrum and multiple coherence peaks instead of one **I don't understand this part so much, do we have any plot associated to this?**. Although our approach is generalizable to the frequency domain, we computed the mutual information between the amplitude co-modulation and phase differences between the broadband signals, for the specific application to the Stuart-Landau model. Note that the Stuart-Landau model already produces complex-valued signals, so a Hilbert-based transformation to obtain analytical signals is not required. The total mutual information was then decomposed into unique, redundant and synergistic terms using PID as described in the Methods.

We simulated a two-node SL network model with unidirectional coupling $g(t)$ from X_i to X_j (Fig. 3A and B). The total mutual information between the joint amplitude co-modulation and phase difference with task variable displayed a clear peak that paralleled the modulation in coupling strength (green curve in Fig. 3C). Most of the interdependence was carried by the modulation in amplitude (orange curve in Fig. 3C). Given that modulation of $g(t)$ drives both amplitude modulation (AM) and phase modulation (PM) in a SL model, we expected the encoding of coupling strength to produce synergistic and redundant effects. Since our approach allows the decomposition of total mutual information in redundant and synergistic terms, we quantified each contribution separately. Indeed, information decomposition showed that a large component is present synergistically in amplitude co-modulation (orange curve in Fig. 3D), with an additional

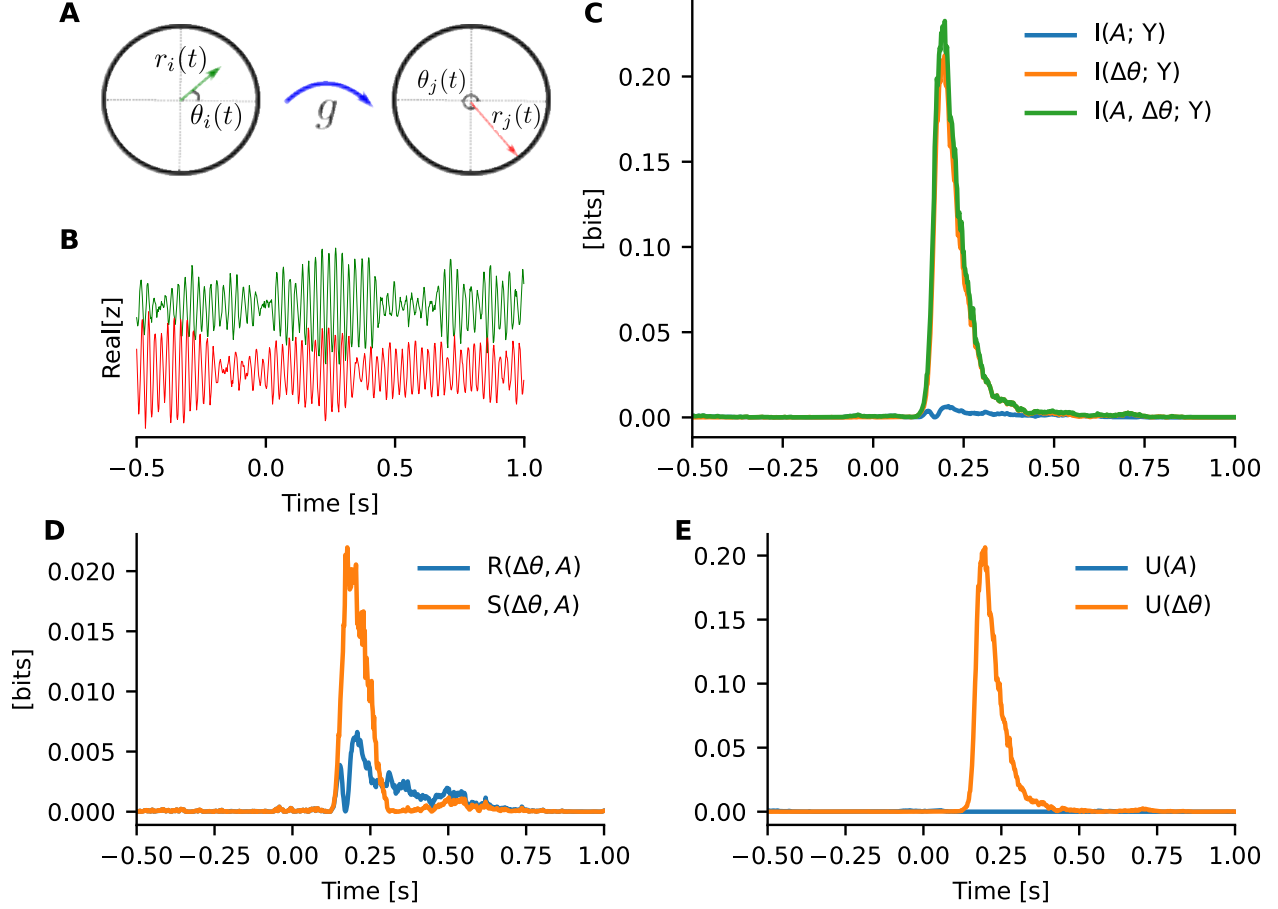


Figure 3: Phase vs. amplitude information in the Hopf model. (A) Simulation of two Stuart-Landau oscillators (See Eq. 18; using the same parameters for both nodes $a = -5$; $\omega = 2\pi 40$ Hz; $\eta = 0.0001$; and coupling strength $C_{12} = 1$) coupled via a unidirectional link from X_i to X_j . The coupling strength g among nodes was transiently modulated from 0 to 0.4 ms according to $g(t) = 8Y/(2\pi f)$, where Y was the maximum coupling amplitude which was increased from 1 to 100 across trials ($n = 5000$). (B) Real part of the ($\text{Real}(Z)$) time-series for the simulated SL system. (C) Stimulus-related information contained in the broadband joint-amplitude (blue; $A = \text{abs}(Z_i Z_j^*)$); the phase-difference (orange; $\Delta\theta = \arg(Z_i Z_j^*)$), and both (green). (D) The redundant (blue) and synergistic (green) information about the stimulus contained in the phase-difference and the joint-amplitude. (E) The unique information about the stimulus in the phase-difference (orange) and the joint-amplitude (blue).

contribution of redundant information (blue curve in Fig. 3D). Given that most of the mutual information among variables is due to the coupling between amplitude co-modulation, we also observed that the unique information in amplitude dominated (orange curve in Fig. 3E), whereas the unique information in phase difference was null (blue curve in Fig. 3E). Overall, the simulation showed that the contribution of amplitude co-modulations and phase relations between oscillatory nodes can be disentangled and quantified in terms of redundant and synergistic contributions.

Information about cognitive variables in phase relations and amplitude co-modulations in whole-brain cortical network model

TO DO Figure 4. The labels V1-V2 and V1-24c should be inverted. It is not clear the definition of two-plets and three-plets. I would remove them. Maybe adjust the colors of the curves.

We next studied the encoding of task variables in whole-brain cortical network oscillatory models using empirically estimated inter-areal coupling strengths. Our goal was to investigate whether we could quantify information contained in amplitude co-modulations and phase relations in more realistic neural oscillatory networks with multiple interacting elements whose interactions are constrained by anatomical data, and to isolate redundant and synergistic terms for both pairwise and higher-order oscillatory interactions. To do so, we created a model of the macaque cortex by simulating a network of coupled Stuart Landau oscillators. We adopted the approach proposed in Joglekar et al. (2018), assuming that the anatomical connectivity between cortical areas is given by the fraction of labeled neurons (FLN) (Markov et al., 2013), which provides a quantitative map of cortical projections (Fig. 4A). The time series display oscillatory patterns across the nodes of the network and a modulation following modulation of coupling strength in the whole network (Fig. 4B). As a reminder, the modulation of the coupling strength reproduced a task-related dependence of varying amplitude across trials and time, similarly to previous simulations. We first investigated how information about task variable was contained at the level of amplitude co-modulations A_{ij} and phase differences $\Delta\theta_{ij}$ between pairs of nodes. To do so, we focused on two brain areas that are known to be strongly anatomically connected: the primary and secondary visual cortices (areas V1 and V2, respectively). We observed that the total (blue curve) and marginal (green and orange curves) mutual information successfully tracked the dynamics of information encoding (Fig. 4C), thus suggesting that information about cognitive variable was contained in both amplitude and phase relations. To quantify the interaction between amplitude and phase encoding, we computed redundant and synergistic information. The temporal evolution of redundant and synergistic encoding showed that both contributed to the overall interaction, peaking approximately at the same time (Fig. 4D). We then quantified the unique information in amplitudes and phases. We observed a positive contribution for both, but peaking at slightly different time points, suggesting that the information in phase synchrony peaks earlier than that in amplitude co-modulation (Fig. 4E). As a control analysis, we studied two distant brain regions: the primary visual cortex (area V1) and the anterior cingulate sulcus (area 24c). Although distant, we

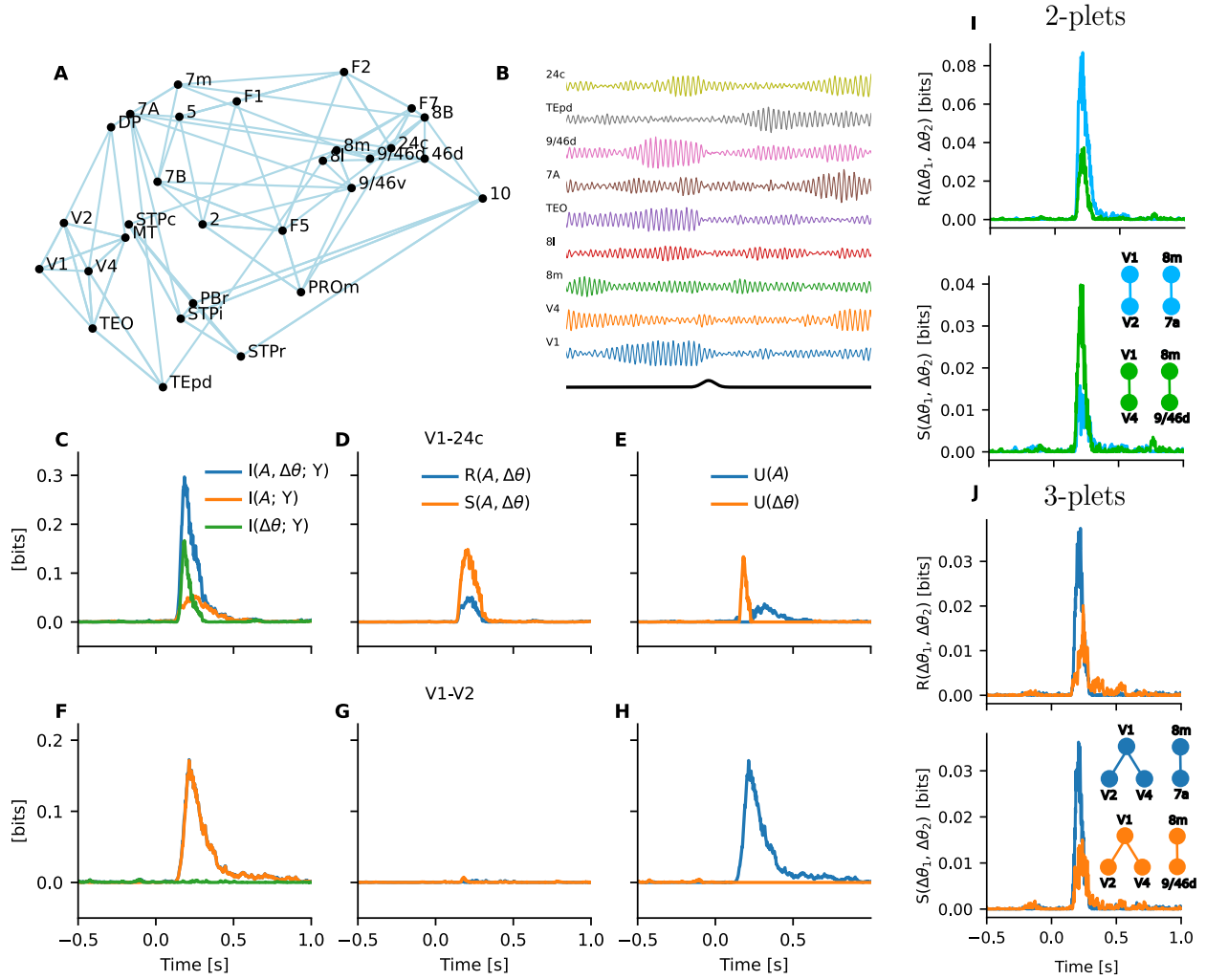


Figure 4: High-order interaction in the SL whole-brain model. (A) Structural connectivity derived from the dataset reported in Markov et al. (2014). Here we show it as a binary network with only the top four connections of each cortical area being drawn (FLN greater than the 87 percentile for each source area). (B) The real part of the ($\text{Real}(Z)$) time-series for the simulated SL whole-brain model for selected 9 cortical areas in one trial. The black time-series on the bottom show the global coupling dynamics $g(t)$. The coupling strength g among nodes was transiently modulated from 0 to 0.4 ms according to $g(t) = 8Y/(2\pi f)$, where Y was the maximum coupling amplitude which was increased from 1 to 100 across trials ($n = 1000$). (C) The time-series of the total mutual information between V1 and 24c contained in their joint amplitude A and phase-difference $\Delta\theta$ about the stimulus (blue curve). Only their amplitude interaction (orange curve), and only their phase-difference (green). (D) The redundancy (blue curve) and synergy (orange curve) between the joint amplitude and phase-difference concerning the stimulus. (E) The unique stimulus information contained in the joint amplitude (orange curve), and only their phase-difference (green). (F-H) The same as C-E but for areas V1 and V2. (I). High-order redundant (top panel) and synergistic (bottom panel) information contained in pairs of edges via phase difference ($\Delta\theta_1$ and $\Delta\theta_2$, respectively). The PID quantities are computed for the phase-difference time-series between nodes $i-j$ and $l-k$. The curves shown are for the two disconnected edges (four nodes in total) with highest synergy and redundancy formed by the edges V1-V2 with 8m-7a (cyan) and V1-V4 and 8m-9/46d (green) (J) Oscillatory higher-order redundant (top panel) and synergistic (bottom panel) information contained in three edges and five nodes.

observed that the total mutual information displayed a strong modulation, therefore suggesting that information about task variable propagates effectively to prefrontal areas (blue curve in Fig. 4F). However, our analysis showed that most of the information was contained in amplitude co-modulations rather than phase relations (orange and green curves in Fig. 4F, respectively). As a confirmation, the redundant and synergistic information in amplitudes and phases were negligible (Fig. 4G), and most was contained in amplitudes in a unique way (Fig. 4H).

We then generalized the analysis of oscillatory interactions beyond pairwise relations, the so-called higher-order interactions. In other words, we investigated whether large-scale oscillatory network encoding task information can recruit more than pairs of nodes. Indeed, we observed oscillatory HOI between pairs of edges displaying redundant and synergistic information about task modulation via phase relations. The results showed that higher-order interactions between two edges between four nodes contained significant information via coupled phase relations (Fig. 4I). This can be interpreted as a correlate of distributed information about cognitive variables via collective phase relations. Finally, we investigated higher-order information between three edges. The results showed that several multiplets displayed non-zero redundant and synergistic information (Fig. 4J).

Overall, these results show that our approach is able to disentangle information about task variables contained in amplitude and phase relations between pairs of nodes in a whole-brain oscillatory model. In addition, these results provide evidence that distributed information about cognitive variables via phase relations between nodes beyond pairwise relations can be extracted from oscillatory activity.

Distributed information in cortical oscillatory networks during working memory

Finally, we aimed to provide proof-of-concept using eletrophysiological data recorded from non-human primates. We analyzed local field potentials (LFPs) simultaneously recorded from lateral prefrontal and posterior parietal cortical areas. Previous studies have shown that LFPs display content-specific beta-band (15–30 Hz) coherence during working memory tasks (Salazar et al., 2012). We first computed the total mutual information between stimulus type and amplitude and phase relations for two LFPs recorded in the frontal (area 8B, located in the dorsolateral prefrontal cortex, anterior to the frontal eye fields, area 8) and parietal (area PE, located in the dorsal portion of the superior parietal lobule) regions. We observed a stimulus-dependent modulation of total mutual information after stimulus onset and during the delay period of the task (Fig. 5A). The total mutual information was decomposed into unique contributions due to amplitude and phase relations. For the session under investigation, the results show that most of the information about stimulus is contained in amplitude co-modulations (Fig. 5C), rather than phase relations (Fig. 5C).

Next, we investigated the presence of higher-order oscillatory interactions from LFPs. To do so, we analyzed two pairs of LFPs recorded from the fronto-parietal network. The first pair included channels located in the caudal part of the inferior parietal lobule (area PG) and parietal area PE,

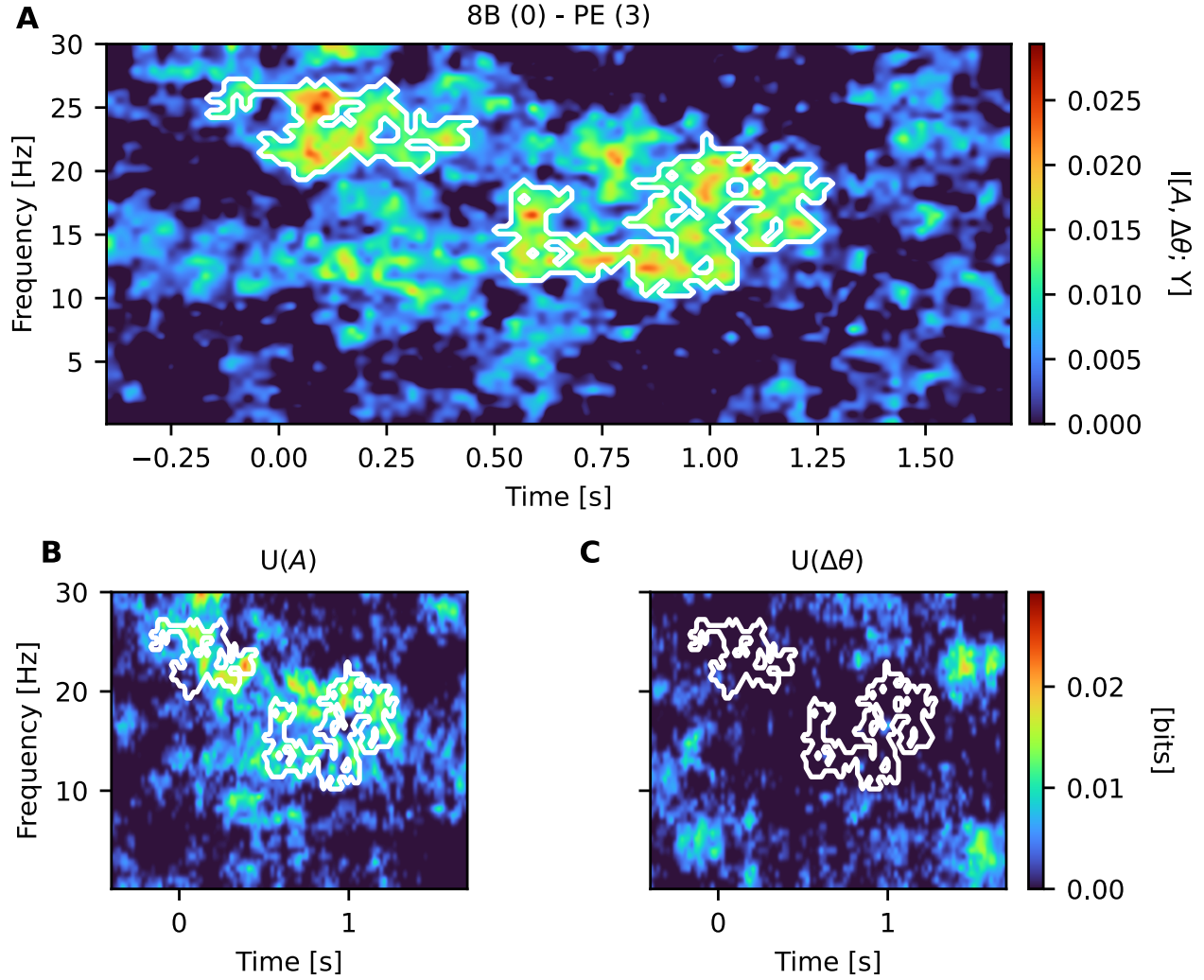


Figure 5: Phase- and amplitude information in fronto-parietal recordings during working memory. **A** Stimulus-related coherence given by the total mutual-information between phase- and amplitude about the stimulus (Eq. 3) for a fronto-parietal channel pair (8B - PE). **B,C** Unique information about the stimulus contained in the joint amplitude and phase-difference components of the signal. Areas delimited by the white line correspond to significant effects obtained via shuffling surrogates ($n = 1000$; $p < 0.05$; corrected using cluster statistics).

respectively. The second pair included channels located in the dorsomedial frontal cortex, adjacent to area 8A and parietal area PE. (Fig. 6A) shows the total mutual information between amplitude co-modulations and phase relations, and stimulus type, showing clear higher-order information structure. We then investigated whether the observed mutual information was due to redundant or synergistic interactions. The results suggest that both redundant (Fig. 6B) and synergistic (Fig. 6C) interactions between higher-order phase relations are present. Overall, we provided proof-of-concept that our methodology can be successfully used to disentangle information contained in amplitude and phase relations in oscillatory networks, and to identify distributed oscillatory higher-order interactions.

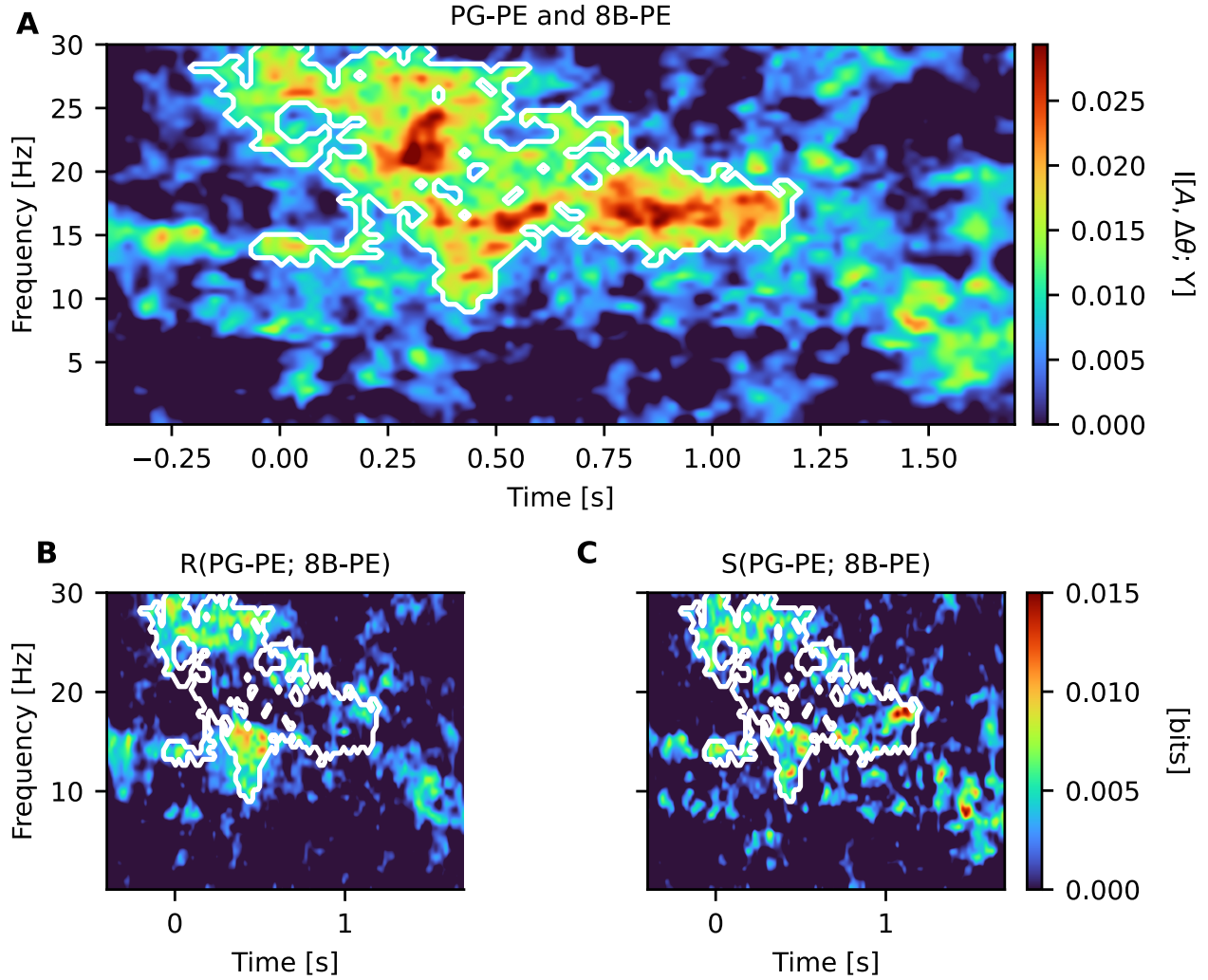


Figure 6: **High-order phase- and amplitude information during working memory.** (A) Stimulus-related coherence given by the total mutual-information between phase- and amplitude edges for parietal-parietal channel pair PG-PE and fronto-parietal channel pair 8B-PE about the stimulus (Eq. 3). The amplitude-phase edge for each channel pair is given by Eqs. 10. (B) High-order redundant information about the stimulus contained in the joint amplitude and phase-difference components of the signal. Areas delimited by the white line correspond to significant effects obtained via shuffling surrogates ($n = 1000$; $p < 0.05$; corrected using cluster statistics). (C) The same as (B) but showing the high-order synergistic information.

DISCUSSION

A large body of evidence supports the view that cognitive functions emerge from the coordinated activity of neural populations distributed across large-scale brain networks (Varela et al., 2001a; Bressler and Menon, 2010), as well as from network-wide, self-organized patterns of information routing (Deco et al., 2015; Battaglia and Brovelli, 2020). However, key questions concerning the underlying neurophysiological mechanisms and computational objectives remain unresolved. For example, how do brain interactions regulate segregation and integration processes to support cognitive functions (Sporns, 2013; Braun, 2015; Deco et al., 2015; Cohen and D’Esposito, 2016; Shine, 2016; Finc, 2020; Wang et al., 2021)? How do functional interactions based on oscillatory synchronization (Engel et al., 2001; Buzsáki and Draguhn, 2004; Fries, 2015) or resonance phenomena (Vinck et al., 2023) carry information about cognitive variables? Do aperiodic transient dynamics and oscillatory phase relations constitute complementary substrates of inter-areal communication and information encoding? How does the brain trade off redundant and synergistic encoding? How do higher-order oscillatory interactions allow distributed encoding over large-scale brain circuits? Progress towards an integrated knowledge in this domain has been limited by the lack of computational tools. In the current study, we provide a novel approach that allows the quantification of information about cognitive processes in neural oscillatory networks.

Spectral Partial Information Decomposition for oscillatory networks

COMMENT: I don’t know if it’s worth presenting this as a new approach called SPID or simply present it a combination of multiple techniques all glued together. **I think that if the approach has not yet been named it can make it things simpler for us to refer to it in the manuscript**

The first contribution of the study consists of having developed a novel approach by combining different information-theoretical and signal processing approaches into a coherence approach for the decomposition of information about continuous or discrete variables (e.g., task variables) contained in interacting oscillatory time series. More specifically, the proposed approach, termed Spectral Partial Information Decomposition (SPID), quantifies task-related information in the frequency domain contained in oscillatory interactions. SPID integrates frequency-domain mutual information (Malladi et al., 2018; Brillinger and Guha, 2007; Young et al., 2020) with the Partial Information Decomposition (PID) framework (Williams and Beer, 2010; Wibral et al., 2017). This integration yields a set of formal rules for decomposing multivariate mutual information between a set of predictors (i.e., brain signals) and a target variable (e.g., a task variable), and defines unique, redundant, and synergistic information components for oscillatory signals. The formalism is general and may be applicable to other domains involving oscillatory complex systems. In the present study, focusing on neural oscillations, we adopt the *minimum mutual information* (MMI) PID, which provides accurate estimates for a broad class of systems consistent with a multivariate Gaussian distribution (Barrett, 2015). In addition, mutual information and entropy are estimated

using Gaussian Copula Mutual Information (GCMi) (Ince, 2017), a semi-parametric, binning-free method that is computationally efficient for the short time series typically encountered in neurophysiological data.

Phase relations and amplitude co-modulations as complementary and interrelated coding schemes

We first validated our approach through simulations based on the Kuramoto models of coupled oscillators, which is a classical framework for studying oscillatory networks. The analysis of the Kuramoto-based simulations showed that SPID can capture the amount of information contained in phase relations between nodes about a task-related variable which modulates the coupling strength (Fig. 1). The time-frequency representation of the mutual information between the phase difference of two nodes (Fig. 1F) provides an additional explanatory dimension with respect to the standard spectral coherence (Fig. 1E), which primarily reflects task-unrelated synchronization. The comparison between spectral mutual information and coherence therefore provides a way for assessing the functional role of phase synchronization in relation to cognitive demands and task manipulations.

We then tested the SPID approach on simulations based on Stuart–Landau models of coupled oscillators, which extends the Kuramoto model by incorporating both phase and amplitude, thus providing a minimal system capable of exhibiting subcritical Hopf bifurcations, as well as transient and steady-state oscillations (Ponce-Alvarez and Deco, 2024). In the simulations based on a pairwise SL model (Fig. 3A), we observed that amplitude co-modulation carries the dominant fraction of task-related information (Fig. 3C and E). However, we also observed that phase and power can interact both redundantly and synergistically (Fig. 3D), suggesting that the degree of interdependence between the two coding schemes can be quantified. Crucially, the SPID analysis revealed a synergistic interaction between amplitude and phase (Fig. 3D), suggesting that task-related information is not merely present in parallel in these two channels, but it can partly emerge from their joint co-variation in ways not predictable from either alone. In more biologically realistic Stuart–Landau model with multiple oscillators coupled according to macaque monkey structural connectivity (Markov et al., 2014), we observed that inter-areal interactions through synchronization and modulation of oscillatory power can carry complementary unique information (Fig. 4C and E). For some pairs of areas, the coupling modulation was primarily driven by amplitude co-modulation, with phase relations providing no unique informative contribution (Fig. 4F and H). However, we also observed that phase relations and amplitude co-modulations contain redundant and synergistic information about task variables (Fig. 4D), or be independent (Fig. 4G).

The whole-brain Stuart–Landau network model (Fig. 4) suggests a distance-dependent pattern: proximal, strongly connected areas (V1–V2) tend to show mixed phase–amplitude information, whereas distant regions (V1–area 24c) showed predominantly amplitude-based information with

negligible phase contribution. This may suggest the presence of a gradient constrained by long-range projections, which may reduce phase relations due to long conduction delays and thus limiting precise phase locking across distant regions (Buzsáki and Draguhn, 2004; Bastos et al., 2015). Under these conditions, amplitude envelope correlations, being more robust to latency variability, may support long-range information transfer, as observed in large-scale MEG and SEEG studies (Combrisson et al., 2025; Brovelli et al., 2015; Combrisson et al., 2024)). Nevertheless, higher-order analyses (Figs. 4I-J) showed that long-range interactions based on phase-relations can be observed in the whole-brain model, and indicating that distributed cortical representations cannot be reduced to amplitude com-modulations only.

The presence of synergistic information implies that treating phase synchrony and power modulations as independent proxies for neural coordination (Lachaux et al., 1999; Siegel et al., 2012) may underestimate the information available in their interdependence. More broadly, this finding is consistent with recent theoretical proposals that aperiodic transient dynamics and oscillatory phase relations constitute complementary substrates of inter-areal communication (Vinck et al., 2023).

Taken together, a central finding across our simulations was that amplitude co-modulation and phase relations can provide complementary encoding schemes of task variables, whose contributions can be disentangled and quantified using our approach. The quantification of the degree of redundant, synergistic and unique information present in each coding scheme, may be used to test hypotheses about the role of inter-areal interactions via either oscillatory synchronization (Engel et al., 2001; Buzsáki and Draguhn, 2004; Fries, 2015) or transients resonance phenomena (Vinck et al., 2025) in cognition. In addition, our approach may profit theoretical development of the Communication Through Coherence (CTC) hypothesis (Fries, 2005, 2015), which posits that phase synchrony between neural populations is the primary mechanism for routing and encoding cognitive information. Future analyses using our approach may provide a nuanced perspective on this claim.

Distributed and higher-order information about task variables via redundant and synergistic oscillatory interactions

A notable contribution of the study is the possibility to investigate the role of distributed and higher-order oscillatory interactions in the structure of information about task variables via redundant and synergistic interactions. We first simulated modulation of neuronal interactions through changes in neuronal synchronization and coupling strength in a network with three nodes using a network of Kuramoto oscillators (Fig. 2). We used different topological organizations of an oscillatory circuit to determine whether information is reflected redundantly or synergistically across interacting edges. A feedforward chain architecture produces predominantly redundant encoding (Fig. 2A), while a convergent collider structure produces synergistic encoding (Fig. 2C). These simulations may provide a direct link between graph-theoretic descriptions of connectivity

motifs and information-theoretic characterizations of neural coding. A predominant redundant encoding, as produced by chain architectures resembling feedforward sensory hierarchies, may confer robustness against signal degradation. This may be identified as a predominant redundant information contained in pairs of edges interacting via phase relations, which would reflect a higher-order interactions between three areas (Fig. 2B). The computational advantage of such coding scheme would be that information about cognitive variable propagates across multiple downstream pathways, so to ensures that no single pathway failure eliminates the representation. In addition, our simulations showed that synergistic encoding, however, as produced by convergence topology (Fig. 2C), was also detectable as a predominant synergistic information in the time-frequency representation (Fig. 2D). A predominant synergistic encoding, as produced by converging architectures resembling top-down influence, may allow functional integration and flexibility properties to the network (Voges et al., 2024). Nevertheless, in anatomically realistic three-node circuits modeled on the V1-V2-V4 circuit, we observed a balanced mixture of both encoding strategies (Fig. 2E-F). In more biologically realistic Stuart-Landau model with multiple oscillators coupled according to macaque monkey structural connectivity, we observed that oscillatory HOI between pairs of edges display both redundant and synergistic information about task modulation via phase relations (Fig. 4I and J), which can be interpreted as a correlate of distributed information about cognitive variables via collective phase relations.

Overall, these results provide evidence that distributed information about cognitive variables via phase relations between multiple brain areas can be extracted from oscillatory activity, and exploit a balance between redundant and synergistic coding schemes. Indeed, the balance between these coding schemes may underpin large-scale robustness and resilience in oscillatory brain network. Synergistic interactions may confer functional advantages over redundancy and unique information by enabling combinatorial coding across brain regions, potentially scaling exponentially with system size Rosas et al. (2020). In contrast, redundancy may support robustness by ensuring that information remains available despite perturbations, at the cost of over-representation, whereas synergy is more vulnerable to disruptions affecting constituent nodes (Mediano, 2022). This would be consistent with the functional advantages of combining robustness and flexibility in distributed cortical circuits (Luppi et al., 2022; Combrisson et al., 2024). More generally, we suggest that the study of the trade-off between redundant and synergistic information about cognitive variables in oscillatory networks may provide a better understanding of the mechanisms regulating segregation and integration in cognition (Sporns, 2013; Braun, 2015; Deco et al., 2015; Cohen and D’Esposito, 2016; Shine, 2016; Finc, 2020; Wang, 2021). To conclude, our study showed that the SPID approach can be used as a tool not only for detecting the presence of higher-order oscillatory interactions, but for disentangling their computational function in terms of collective encoding strategies.

Proof-of-concept on LFPs from fronto-parietal network during working memory

Finally, we provided a proof-of-concept on local field potentials (LFPs) recorded from macaque parietal and premotor cortical regions during a working memory task. We searched for information carried by beta oscillations (15-30Hz) about the stimulus identify during delayed-matched-to-sample task. For an exemplar session, we observed a stimulus-dependent modulation of total mutual information after stimulus onset and during the delay period of the task (Fig. 5A), thus confirming previous findings showing content-specific beta-band fronto-parietal synchronization during visual working memory (Salazar et al., 2012). The total mutual information was decomposed into unique contributions due to amplitude and phase relations. For the session under investigation, the results show that most of the information about stimulus was contained in amplitude co-modulations (Fig. 5B), rather than phase relations (Fig. 5C). This suggests that future studies may profit from disentangling synchrony- and amplitude-based information components.

The higher-order analysis (Fig. 6) further reveals that pairs of fronto-parietal recording sites jointly reflect stimulus information through both redundant and synergistic interactions during the memory delay period. The synergistic component is particularly noteworthy: it implies that certain aspects of the memorized stimulus are only jointly contained by multiple fronto-parietal channel pairs simultaneously. However, the result confirm the prediction from whole-brain simulations, suggesting a balanced encoding based on redundancy and synergistic information. The co-existence of redundant and synergistic higher-order encoding within the same fronto-parietal network suggests that the same oscillatory system simultaneously supports reliable broadcast and flexible integration of working memory content, an organizational principle with direct implications for biologically plausible models of working memory maintenance (Dotson et al., 2018; Miller et al., 2018).

Methodological considerations

Recent advances in information theory have provided several tools to characterize high-order interactions (HOIs) in the frequency domain and they have been applied to complex physiological networks (Faes et al., 2022; Sparacino et al., 2023; Antonacci et al., 2025). SPID addresses a fundamentally complementary question: rather than quantifying the nature of oscillatory interactions, it quantifies the information about external task or cognitive variables contained in the interactions between signals. This distinction maps onto the difference between functional connectivity (signal-to-signal statistical relations) and information encoding (brain-behavior dependence). An important direction for future work is to integrate these two perspectives, combining the study of multivariate information in the frequency domain in complex oscillatory systems with the task-variable decomposition of SPID, so to obtain a unified account of how intrinsic oscillatory interactions are reflected in external task variables. An additional potential extension is the study of across-orders interactions from pairwise to higher-order interactions in oscillatory networks, with the potential application to multiscale systems.

Potential limitations

Several aspects of the current implementation merit consideration. First, the minimum mutual information (MMI) PID estimator (Barrett, 2015), while computationally tractable and well-calibrated for systems with multivariate Gaussian statistics, provides only a lower bound on synergistic information and may not faithfully decompose the unique information atoms across all probability regimes (Williams and Beer, 2010; Wibral et al., 2017). Future implementations could benefit from more principled lattice-based PID comparing different approaches for information decomposing (Liardi et al., 2026). Second, the Gaussian-copula mutual information estimator (Ince, 2017) provides a robust and sample-efficient approximation for short neural data series, but its reliance on rank-based normalization to Gaussian may underestimate information contained in non-monotonic relationships between signal features and task variables. Finally, the neurophysiological results reported here are based on a single session from a single animal, and therefore constitute a proof-of-concept; validation across multiple animals, sessions, and task designs is necessary to establish the generality of encoding principles.

Future directions

Several natural extensions of SPID are motivated by the present results. First, SPID as formulated here is symmetric and does not provide directed estimates of which oscillatory signal drives encoding in another. A possible future direction is to incorporate directionality analyses based on the Wiener-Granger causality principle to identify asymmetric interactions in oscillatory networks. Second, extending SPID to cross-frequency interactions, such as theta-phase to gamma-amplitude coupling in hippocampo-cortical circuits, would address whether multiplexed frequency codes carry unique or synergistic task information and could clarify the computational role of cross-frequency nesting in cognition.

Conclusions

Spectral Partial Information Decomposition provides a unified approach for characterizing how oscillatory neural networks reflect cognitive information in the frequency domain. By revealing the complementary and synergistic roles of phase synchrony and amplitude co-modulation, and extending the characterization of redundant and synergistic encoding to higher-order interactions, SPID substantially expands the theoretical and empirical toolkit for investigating neural coding in distributed oscillatory networks. [Furthermore, its formulation is highly general, allowing easy adaptation of both the mutual information estimators used and the phenomena under investigation, such as cross-frequency encoding and interactions across different orders.](#) Here, we show that systematic validation across canonical oscillatory models, biologically realistic whole-brain simulations, and beta-band oscillations in the fronto-parietal network establishes SPID as a method for probing how information about cognitive variables is reflected in distributed oscillatory brain networks.

MATERIALS AND METHODS

Spectral Partial Information Decomposition for task-related oscillatory networks

We propose a novel approach called Spectral Partial Information Decomposition (SPID) that quantifies task-related information contained in oscillatory interactions in the frequency domain. SPID can be used to disentangle information about task variables contained in modulations in power or phase difference between oscillatory signals. In addition to pairwise relations, SPID can quantify redundant and synergistic information contained in higher-order oscillatory interactions in the frequency domain. SPID is based on mutual information in frequency domain and partial information decomposition framework, whose specific applications to the analysis of neural interactions will be reviewed in the following sections.

Mutual information in frequency domain between neural signals

A standard measure of the statistical dependence between two random variables is Shannon’s mutual information. As a reminder, mutual information is defined as:

$$I(X_1; X_2) = H(X_1) - H(X_1|X_2) = H(X_1) + H(X_2) - H(X_1, X_2) \quad (1)$$

Variables X_1 and X_2 may represent the neural activity of two brain areas or neural populations, respectively. $H(X_1)$ is the entropy of X_1 , and $H(X_1|X_2)$ is the conditional entropy of X_1 given X_2 . Mutual information can be extended to quantify the statistical dependence between X_1 and X_2 in the frequency domain. This is achieved by decomposing the signals into their frequency components using Fourier or wavelet transforms, yielding a set of complex coefficients. Each coefficient can be represented in polar form as $X(f) = r(f)e^{i\theta(f)}$, where $r(f)$ denotes the amplitude and $\theta(f)$ the phase of the signal at frequency f . Alternatively, the spectral decomposition can be expressed in Cartesian form as $X(f) = a(f) + ib(f)$, where $a(f)$ and $b(f)$ represent the real and imaginary parts of the signal, respectively. Mutual information in frequency (MIF) is defined as the Shannon mutual information between the Fourier coefficients of two stochastic processes, X_1 and X_2 at a given frequency f (Malladi et al., 2018; Brillinger and Guha, 2007; Young et al., 2020). In practice, MIF computed as the mutual information between the real and imaginary components of those coefficients:

$$I_f(X_1; X_2) = H(a_1(f), b_1(f)) - H(a_1(f), b_1(f) | a_2(f), b_2(f)), \quad (2)$$

where $H(x, y)$ denotes the joint entropy. Mutual information in frequency (MIF) captures the statistical dependence between two signals specifically associated with oscillatory activity at frequency f . We should note that mutual information in the frequency domain and spectral coherence are closely related Bernardi and Lindner (2015); however, unlike coherence or traditional cross-frequency coupling metrics, MIF quantifies dependencies arising from both linear and nonlinear relation-

ships. When the time series are linear Gaussian processes, coherence has a direct, well-defined relationship to MIF (Malladi et al., 2018; Young et al., 2020). In this case, MIF becomes a monotonic transformation of the coherence magnitude, linking information-theoretic and spectral measures of interdependence. Thus, MIF extends beyond linear dependencies to capture a broader range of frequency-specific coupling and information exchange between neural signals.

The same approach can be used to compute the mutual information between the Fourier coefficient of a single neural signal X_1 and a task-related variable Y , which may represent the trial-by-trial evolution of stimulus properties, decisions, or internal cognitive variables. In this case, the task-related MIF is defined as

$$I_f(X_1; Y) = H(a_1(f), b_1(f)) - H(a_1(f), b_1(f) | Y),$$

and it quantifies how much information about Y is contained in the complex spectral representation of X_1 at frequency f . Intuitively, $I_f(X_1; Y)$ measures the reduction in uncertainty about the amplitude and phase of the neural oscillation at frequency f when the task variable Y is known. In other words, it captures frequency-specific components of neural activity that systematically vary with the task or behavioral condition. Mathematically, large values of $I_f(X_1; Y)$ indicate that the distribution of the Fourier coefficients of X_1 differs across levels or values of Y , reflecting a task-modulated change in the spectral structure of the signal. This formulation thus provides an information-theoretic framework for identifying frequency bands that contain behaviorally or cognitively relevant information.

Partial Information Decomposition for pairwise and higher-order functional neural interactions

The second building block of our approach is Partial Information Decomposition (PID). PID provides a mathematical framework for the decomposition of multivariate mutual information between a system of predictors and a target variable into unique, redundant and synergistic information terms (Williams and Beer, 2010; Wibral et al., 2017). PID can be used to decompose the information carried by a set of neural signals X_1 and X_2 about a task-related variable Y , which may represent the trial-by-trial evolution of stimulus properties, decision or internal variables. When only one signal, X_1 or X_2 , individually carries information about a target variable Y , this is termed *unique* information - the information that is specific to each source about Y . If both X_1 and X_2 convey the same information about Y , this shared component is called *redundant* information. Conversely, if neither variable alone provides information about Y , but their joint observation does, this corresponds to *synergistic* information. Redundant and synergistic components thus quantify how interactions between neural signals contribute to the information about Y . Analytically, the PID for the three-variables case - i.e., two source variables case - decomposes the total mutual information between X_1 and X_2 with a target variable Y as:

$$I(X_1, X_2; Y) = U(X_1; Y) + U(X_2; Y) + R(X_1, X_2; Y) + S(X_1, X_2; Y) \quad (3)$$

where $U(X_1; Y)$ and $U(X_2; Y)$ are the unique information carried by the two areas, respectively, and $R(X_1, X_2; Y)$ and $S(X_1, X_2; Y)$ are the redundancy and synergy terms, respectively. In addition, the PID formulation links to classical Shannon measures of mutual information as:

$$I(X_1; Y) = U(X_1; Y) + R(X_1, X_2; Y) \quad (4)$$

$$I(X_2; Y) = U(X_2; Y) + R(X_1, X_2; Y) \quad (5)$$

The governing equations form an under-determined system: only three quantities can be computed from the data (i.e., the mutual information quantities $I(X_1, X_2; Y)$, $I(X_1; Y)$, and $I(X_2; Y)$) for the four terms of the PID (i.e., two unique information terms, redundancy, and synergy). Here, we exploit the so-called *minimum mutual information* (MMI) PID, which has been shown to provide correct estimations for a broad class of systems following a multivariate Gaussian distribution (Barrett, 2015). According to the MMI PID, redundant information carried by pairs of brain regions is given by the minimum of the information provided by each individual source to the target:

$$R_{\text{MMI}}(X_1, X_2; Y) = \min [I(X_1; Y), I(X_2; Y)] \quad (6)$$

Then, synergistic information can be computed by substituting Eqs. 4, 5, and 6 into Eq. 3 and rearranging the terms:

$$S_{\text{MMI}}(X_1, X_2; Y) = I(X_1, X_2; Y) - I(X_1; Y) - I(X_2; Y) + R_{\text{MMI}}(X_1, X_2; Y) \quad (7)$$

By definition of the MMI approach, redundant functional interactions contain the same information about the target variables as that contained in individual brain areas. On the other hand, synergistic connectivity reveals functional interactions that cannot be explained by individual contributions, but only by their combined and collective coordination.

The approach can be generalized to quantify task-related information present in interactions beyond pairwise relations, the so-called higher-order functional interactions (HOI) or correlations. The definitions of redundancy and synergy can be generalized to higher order by simply iterating over all regions of a multiplet. By considering a multiplet as a set of n variables $X_n = \{X_1, \dots, X_n\}$, the redundant information contained in each higher-order multiplet about the target

$$R_{\text{MMI}}(X_n; Y) = \min_i I(X_i; Y) \quad (8)$$

Then, higher-order synergistic information can be computed as:

$$S_{\text{MMI}}(X_n; Y) = I(X_n; Y) - \max_i I(X_i^-; Y) \quad (9)$$

where X_i^- is the set of variables of X_n excluding the source variable i . Equations 11 and 12 represent

the redundant and synergistic information contained in multiplets of brain signals about the target variable Y , respectively.

We used Gaussian-Copula Mutual Information (GCMI) (Ince, 2017), which is a semi-parametric binning-free technique to calculate mutual information and entropy estimates. In the GCMI approach, which includes a rank based normalisation of the data, entropy values are computed using the differential entropy formula for a Gaussian distribution. The transformed data allows entropy estimation based on the determinant of the covariance matrix and the assumption of Gaussian marginals. The GCMI is a robust rank-based approach that allows detecting any type of monotonic relation between variables. More precisely, the GCMI is a lower-bound approximate estimator of mutual information for continuous signals based on a Gaussian copula, and it is of practical importance for the analysis of short and potentially multivariate neural signals. Note, however, that the GCMI does not detect non-monotonic (e.g., parabolic) relations. As an example, this methodology has been applied to investigate the information about learning signals, such as information gain, contained in cortico-cortical co-modulations in broadband high-gamma MEG activity (60-120Hz) (Combrisson et al., 2025).

Frequency-domain measures of task-related information in phase relations

Although consensus has not been reached regarding the mechanisms support neural interactions, most accounts emphasize a central role of co-modulations in aperiodic transient dynamics (Vinck et al., 2023, 2025) and phase relations between neural oscillations (Engel et al., 2001; Fries, 2015; Vezoli et al., 2021; Bastos et al., 2015). Quantifying the information contained in phase relations is therefore crucial.

Here, we propose a methodology to measure task-related information in the frequency domain contained in pairwise and higher-order interactions, and to decompose redundant and synergistic contributions.

To do so, we consider the cross-spectral density between two time-domain signals with Fourier transforms $X_n(f)$ and $X_m(f)$ as $E_{nm}(f) = X_n(f) X_m^*(f)$, where $*$ represents the complex conjugate. It measures the shared spectral power and phase relationship between the two time series. The cross-spectrum $E_{nm}(f) = X_n(f) X_m^*(f)$ is a complex quantity that can be expressed in polar form as $E_{nm}(f) = |E_{nm}(f)| e^{j\phi_{nm}(f)}$.

The magnitude represents the strength of the linear relationship between the signals $x_n(t)$ and $x_m(t)$ at frequency f , and the co-modulation in power. The phase indicates the phase difference between the two signals. Together, these components describe how strongly and in what phase relationship the two signals share energy across frequencies. We thus define the amplitude co-modulations $A_{nm}(f)$ and phase-differences $\Delta\theta_{nm}(f)$ as:

$$\begin{aligned}
A_{nm} &= |X_n(f)X_m^*(f)|; \\
\Delta\theta_{nm} &= \arg(E_{nm})
\end{aligned}
\tag{10}$$

where $*$ corresponds the complex conjugate, and $\arg$ to the argument function.

We then define the total mutual information contained in power co-modulations A_{nm} and phase difference $\Delta\theta_{nm}$ between pairs of brain signals about a task variable Y as $I(A_{nm}, \Delta\theta_{nm}; Y)$. The mutual information in frequency, $I(A_{nm}, \Delta\theta_{nm}; Y)$, quantifies how much information about the task variable Y is jointly carried by the amplitude co-modulations A_{nm} and the phase differences $\Delta\theta_{nm}$ between pairs of signals. In other words, it measures the extent to which task-related changes can be explained by variations in both the shared spectral power and the relative phase relationships of the interacting signals across frequencies. The quantity $I(A_{nm}, \Delta\theta_{nm}; Y)$ can be interpreted as a *task-related coherence measure*, as it quantifies the mutual information between task variable Y and the joint amplitude co-modulations and phase relationships between signals x_n and x_m . It thus reflects how task variable modulates the spectral coupling between pairs of signals.

Task-related coherence, however, conflates contributions from A_{nm} and $\Delta\theta_{nm}$. In the case of spectral coherence, one may estimate the contribution of phase difference solely by computing it after normalizing the spectra and cross-spectra of the signals, which yields the phase-locking value (Rosenblum et al., 2001; Bastos and Schoffelen, 2016). Here, we propose a decomposition based on PID as described above, yielding unique information contained in either amplitude co-modulations A_{nm} and $\Delta\theta_{nm}$, their redundant and synergistic contributions. The redundant information reflects information about the task variable shared by both amplitude co-modulations and phase differences. Whereas, the synergistic atom represents information about the task variable that can be learned by looking at the interaction between A_{nm} and $\Delta\theta_{nm}$. These terms can be computed using the formalism described above.

This notion can be generalized to measure the information about task variables contained in interacting edges via phase differences. The rationale is that pairs of brain regions that share information about task variables through modulations in phase difference may form a network of interacting edges. Such network may track collectively either by means of redundant or synergistic information. This may be measures by considering equations 11 and 12, and using as source variables of the PID the edge-wise phase differences.

By considering a multiplet as a set of k phase differences between pairs of brain signals, $\Delta\theta_k = \{\Delta\theta_{n_1m_1}, \dots, \Delta\theta_{n_km_k}\}$, the redundant information contained in each higher-order multiplet about the target variable is given by:

$$R_{\text{MMI}}(\Delta\theta_k; Y) = \min_i I(\Delta\theta_i; Y) \tag{11}$$

Then, higher-order synergistic information can be computed as:

$$S_{\text{MMI}}(\Delta\theta_n; Y) = I(\Delta\theta_n; Y) - \max_i I(\Delta\theta_{i-}; Y) \quad (12)$$

The redundancy and synergy measures quantify how multiple phase-difference interactions between brain signal collectively track information about a task variable. The redundant information $R_{\text{MMI}}(\Delta\theta_k; Y)$ captures the portion of task-related information that is shared across all phase-difference pairs within a multiplet, that is, information redundantly represented by each edge. In contrast, the synergistic information $S_{\text{MMI}}(\Delta\theta_n; Y)$ reflects information that emerges only when considering the joint activity of multiple edges, beyond what any single edge contributes individually. Together, these measures indicate whether task-related information in a network of interacting edges is organized redundantly (overlapping) or synergistically (complementary) across phase-based interactions.

Computational model and simulation of neural oscillatory interactions

We tested the methodology on simulated data using dynamical system models that captures both amplitude and phase-based interactions. We detail the computational model used to simulate the oscillatory dynamics of the nodes, namely the Kuramoto and the Stuart-Landau model. Next, we describe two network configurations used: the first consisting of two interconnected nodes, and the second based on the macaque anatomical connectivity obtained via retrograde tracing, providing the structural connectivity among nodes.

The Kuramoto network model

The Kuramoto model describes a set of N interconnected nodes with oscillatory dynamics governed by Eq. 13.

$$\frac{d\theta_j}{dt} = \omega_j + g \sum_i C_{ji} \sin(\theta_i - \theta_j) + \eta_j \quad (13)$$

where ω is the natural frequency of the oscillator in rad/s, g regulates the global coupling between nodes, C is the adjacency matrix specifying structural connectivity, and η is a white noise input to the system with zero mean and variance σ^2 . The overall synchrony among nodes is given by the Kuramoto order parameters r and ψ given by Eq. 14.

$$re^{i\psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}, \quad (14)$$

where r captures the phase-coherence of the population of oscillators and ψ their average phase.

687 The Stuart-Landau network model

688 Whereas the Kuramoto model describes the phase dynamics of coupled oscillators assum-
689 ing fixed amplitudes, focusing only on how phases synchronize, the Stuart-Landau (SL) model
690 describes both amplitude and phase dynamics, allowing oscillations to emerge, grow, or decay.
691 Thus, the Kuramoto model is phase-only, while the SL model is a full amplitude-phase oscillator
692 model. The Stuart-Landau model is the canonical normal form of a Hopf bifurcation, which is
693 a fundamental mechanism by which a system transitions from a stable fixed point to oscillatory
694 behavior. In a subcritical Hopf bifurcation, the emerging limit cycle is initially unstable, meaning
695 that small perturbations can cause sudden transitions to large-amplitude oscillations or bistable
696 behavior. In contrast, a supercritical Hopf bifurcation produces a stable limit cycle that grows
697 smoothly from the fixed point as the system parameter changes. The SL model of a single node is
698 given by the Equation 15.

$$\frac{dz}{dt} = (a + i\omega)z - |z|^2z + \eta, \quad (15)$$

699 where z is a complex-valued variable representing the evolution of the dynamical system, ω is the
700 natural frequency of the oscillator in rad/s, a is a control parameter dictating how far the system is
701 from the sub-Hopf bifurcation (Figure 1A), $|z|^2$ is the absolute value of z , and η is a white noise
702 input to the system with mean zero and variance σ^2 . For values of $a < 0$ the system displays
703 damped oscillations; hence, the system $z = 0$ is a spiral or focus point in its state-space. For $a > 0$,
704 the system oscillates, and the stable points are $z = 0$, which correspond to the resting state and
705 $z = \pm\sqrt{a}$ to a limit cycle. The transition between these states happens at the bifurcation at $a = 0$. It
706 is possible to show that the state variable z encapsulates the evolution of the amplitude and phase
707 terms of an oscillator. By writing z as a vector in the complex plane, with length r , and phase θ , i.e.,
708 $z = re^{i\theta}$ and plugging it into Equation 15 with $\eta = 0$, one gets:

$$\frac{dr}{dt} = ar - r^3, \quad (16)$$

$$\frac{d\theta}{dt} = \omega. \quad (17)$$

709 Notice in the differential equations 16 and 17, amplitude and phase components evolve indepen-
710 dently. The fact that the amplitude r and phase θ terms evolve in parallel makes it convenient for
711 studying how cognitive variables are reflected in power modulations or phase relations.

712 Whole brain cortical model based on Stuart-Landau oscillators

713 The Stuart Landau model can be used to simulated whole-brain oscillatory networks by cou-
714 pling nodes via an adjacency matrix C_{ij} representing the structural connectivity between nodes.
715 To determine the inter-regional connectome and the number of projections originating from each
716 cortical area in the whole-brain SL model, we used data reported by Markov et al. (2014), encom-

passing 29 areas from the macaque cerebral cortex. Restricted to these 29 areas, the connectome is edge-complete, i.e. there are no missing inter-regional connections that have not been characterized (Markov et al., 2013). The directed weight from a source to a target area is estimated to be proportional to the Fraction of Labeled Neurons (FLN). The FLN is defined as the number of labeled neurons in a source area divided by the total number of labeled neurons in all source areas. As retrograde tracers are used, the FLN can be considered as an indication of the relative of neurons sending long-range projections from one source area to a target.

The Equation 15 of the full network model is thus given by:

$$\frac{dz_j}{dt} = (a_j + i\omega_j)z_j - |z_j|^2 z_j + g \sum_{k=1}^N C_{jk}(z_k - z_j) + \eta_j, \quad (18)$$

where the indexes j and k (notice that each node is theoretically allowed a distinct intrinsic frequency and a parameter Ponce-Alvarez and Deco (2024)), whereas g is a parameter regulating the global coupling between each node in the network. The coupling strengths are scaled with the hierarchy (h) of each cortical area to amplify inputs arriving to associative areas, yielding:

$$\frac{dz_j}{dt} = (a_j + i\omega_j)z_j - |z_j|^2 z_j + g(1 + \kappa_j h_j) \sum_{k=1}^N C_{jk}(z_k - z_j) + \eta_j, \quad (19)$$

where, h_j is the hierarchical level of node j , and κ_j a parameter regulating the amplification due to hierarchy.

Simulating task-related modulations

Simulation of task-related modulation of the network dynamics and interactions to the Kuramoto and Stuart-Landau simulations was done by modulating the global coupling among nodes. That is, the parameter g in Eqs. 13 and 15 was treated as a dynamic variable $g = g(t)$. More specifically, the coupling strength among nodes was transiently modulated from 0 to 0.4 ms according to $g(t) = 8Y / (2\pi f)$, where Y was the maximum coupling amplitude which was increased from 1 to 100 across trials. When computing information-theory quantities the the maximum coupling amplitudes Y was considered as the continuous stimulus label.

Electrophysiological dataset: distributed neural oscillations during working memory

In order to test our approach on neural data, we analyzed local field potentials (LFPs) recorded simultaneously from the lateral prefrontal and posterior parietal cortical areas, exhibiting content-specific coherence in the beta band (15–30 Hz) during working memory tasks in macaque monkeys (Salazar et al., 2012). Monkeys were trained to perform an oculomotor delayed match-to-sample task, which required them to match either the identity or the position of the sample object. During visual fixation, a sample stimulus, consisting of one of three possible objects, was presented for 0.5 s, followed by a random delay of 0.8–1.2 s. At the end of the delay, a

match stimulus was presented, consisting of the previous sample object (target) and a distractor object at a different position. A saccadic eye movement to the target was rewarded with juice.

Because the number of distinct cues is low ($n=3$), we increased statistical power in our SPID analysis by pooling identity- and position-matching trials, yielding $n=6$ distinct stimulus types. Note that, as a result, the observed encoding effects reflect representations of the task that combine overlapping contributions from both trial types.

Available code resources

All models used for numerical simulations were implemented in Python, relying on scientific computing libraries such as NumPy, SciPy, and JAX. The functions employed for information-theoretic and partial information decomposition (PID) analyses are available in the Frites package (Combrisson et al., 2022) and the HOI package (Neri et al., 2024). The dependencies required to run the simulations, along with five Jupyter notebooks containing step-by-step code to reproduce the figures presented in this manuscript, are provided in the following GitHub repository: github.com/brainets/info-oscillatory-hoi.

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